Lab 4 Report

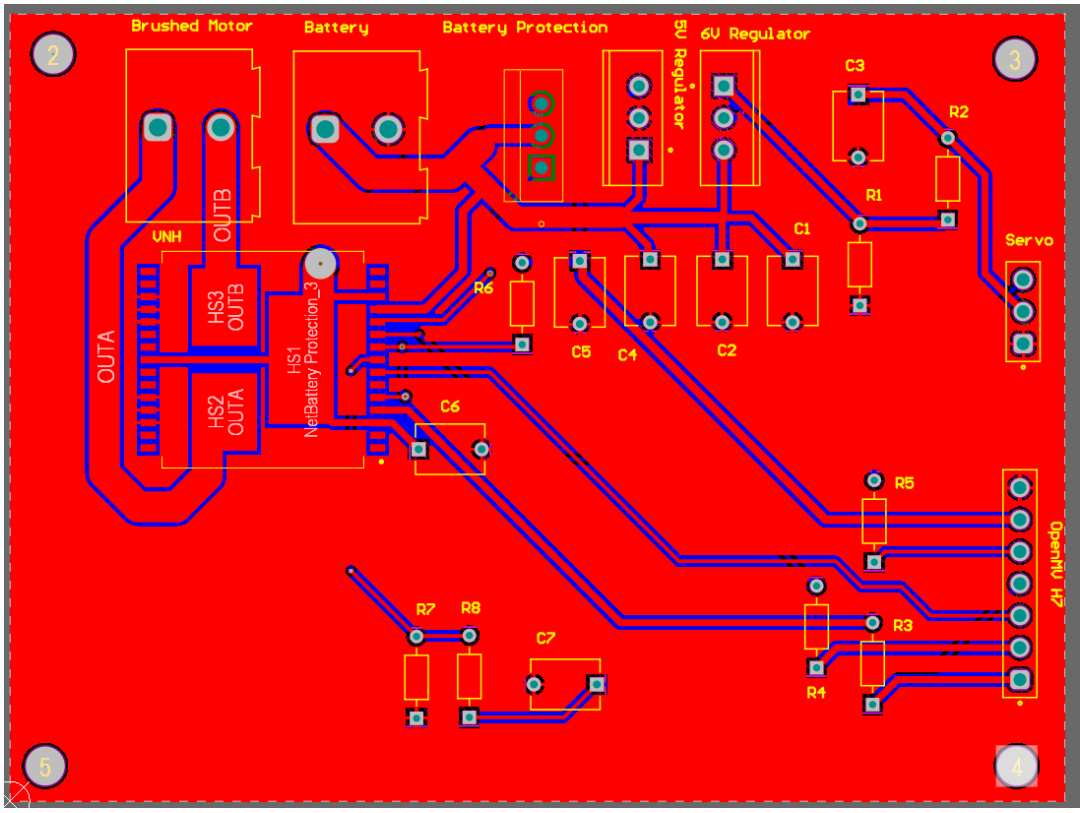
Nicolas Wong-Shea, Temirlan Ryspaev, Xiang Mao, Justin Spenker

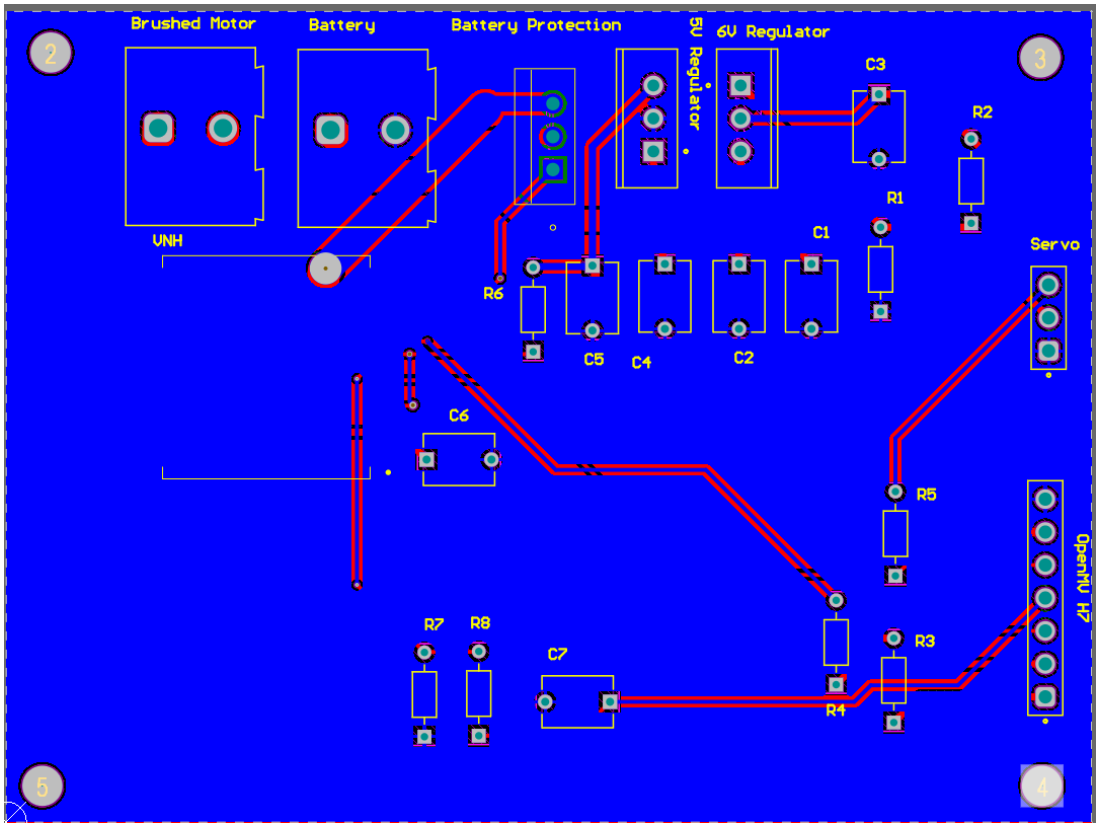
Each group member was responsible for working on a part of the PCB. Every group member contributed in a discussion style on placement, wiring, and parameter adjustments. We came together as a group to troubleshoot and design to create the final design. Everyone contributed roughly equally to the project.

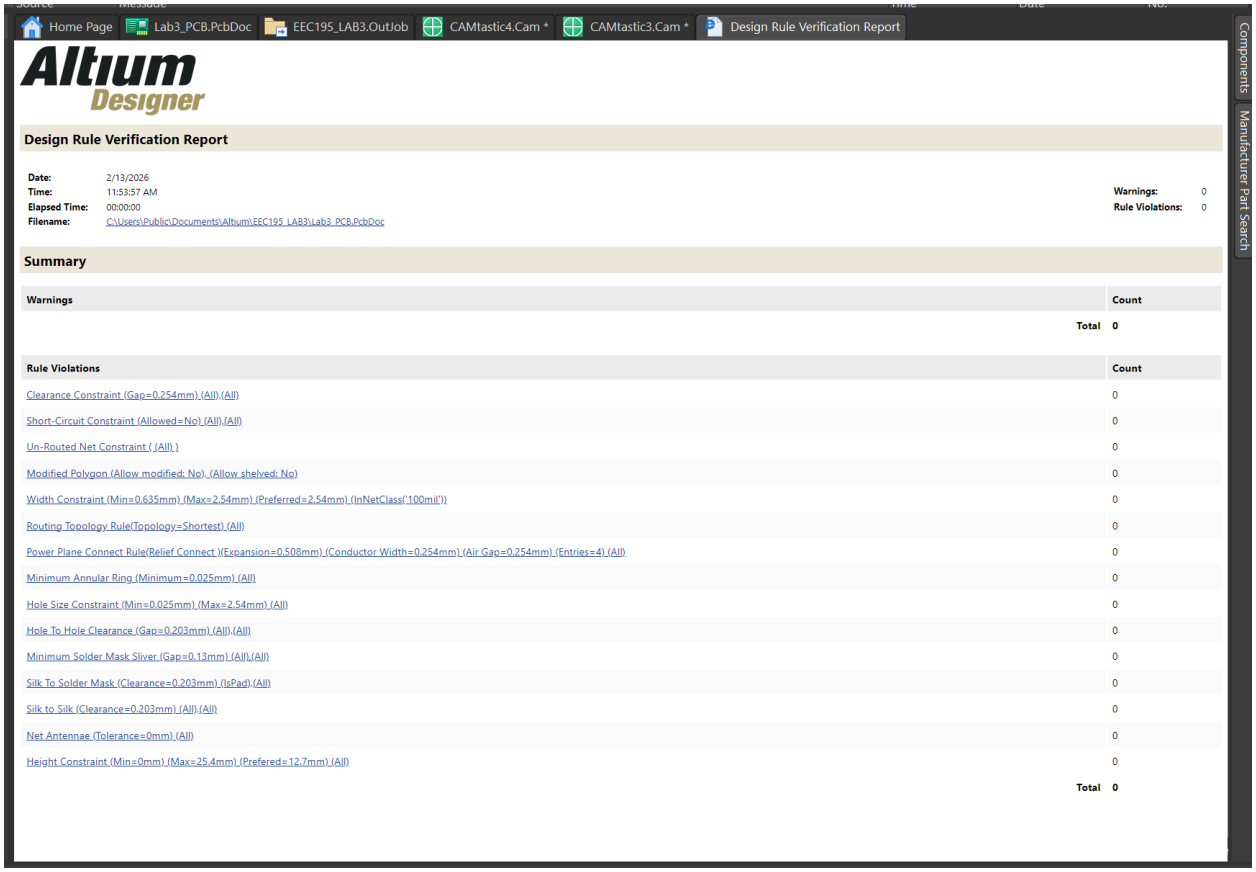
EEC195A Lab #: 4

Names: Justin Spenker
Nicolas Wong-Shea
Temirlan Ryspaev
Xiang Mao

	Score	TA signature / date
PCB	✓	Feb 13
TOTAL LAB SCORE		


Brushed Motor
Battery
Battery Protection
5V Regulator
6V Regulator
C3
R2
R1
C1
Servo
VNH
OUTB
HS3
OUTB
OUTA
HS1
NetBattery Protection_3
HS2
OUTA
R6
C5
C4
C2
C6
R5
OpenMV H7
R7
R8
C7
R3
R4
2
3
5
4


Brushed Motor
Battery
Battery Protection
5V Regulator
6V Regulator
C3
R2
R1
C1
Servo
VNH
R6
C5
C4
C2
C6
R5
OpenMV H7
R7
R8
C7
R3
R4
2
3
5
4



DFM Checks

The minimum distance values from the design data are displayed and referenced against our Standard & Advanced capabilities. Whenever possible, designing according to our Standard capabilities will limit manufacturing costs, lead time, and quality concerns.

	Min. Distance	Bay Area Circuits Standard Capability	Bay Area Circuits Advanced Capability
copper to board edge:	0.0 mil (0 mm)	10 mil	0 mil
copper to plated hole:	12.5 mil (0.318 mm)	10.5 mil	5 mil
copper to non-plated hole:	-	8 mil	4 mil
copper to copper:	10.0 mil (0.254 mm)	6 mil	2 mil
copper ring:	2.5 mil (0.064 mm)	6 mil	1 mil
track width:	15.0 mil (0.381 mm)	6 mil	2 mil

Green = Standard Capability, Orange = Advanced Capability, Red = Not Within Capability

Potential Show Stoppers

Insufficient Annular Ring (10 violations) 1 2 3 4 5

Problems Automatically Fixed

Insufficient Soldermask Clearance (66 violations) 1 2 3 4 5

NOTE: The insufficient annular ring violation can be easily modified to support AdvancedPCB production. I had catered the design for the BayArea PCB. Not a problem for final design.